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Domain & SSL Certificate Expiry Alert System

Architecture Selection

Layered Architecture for the Domain & SSL Certificate Expiry Alert System.

Architectural Choice and Reasons

1. Clear separation of responsibilities:

Layered Architecture separates the system into Presentation, Business/Service, and Data layers. The User Dashboard can handle interaction with the SysAdmin/Security Officer, while the Domain Monitoring, SSL Certificate Monitoring, and Alert & Notification Services handle the main system logic. The Domain & Certificate Database handles data storage. This makes the system easier to understand, develop, and maintain.

2. Easier maintenance and management:

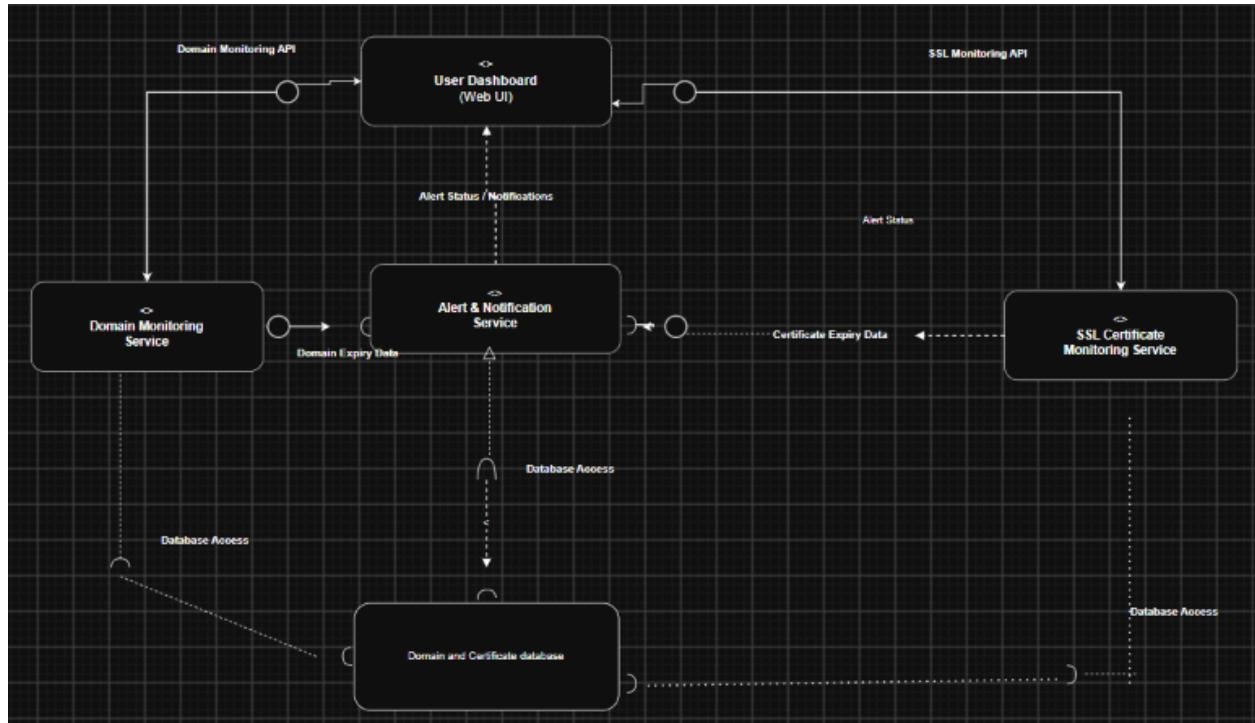
The monitoring and alerting functionality can be organized independently from the user interface and database. For example, changes to the dashboard can be made without changing the database structure. This separation makes it easier to maintain the system as the number of monitored domains and SSL certificates increases.

Security Advantage

Layered Architecture helps improve security by separating the presentation layer from the data layer. The User Dashboard does not directly access the database; requests pass through the business/service layer. This allows access control and validation to be handled before sensitive domain and certificate information is accessed.

Performance Benefit

The monitoring and alerting operations are handled in the Business/Service Layer rather than directly through the user interface. Domain and SSL expiry checks can therefore be processed independently of dashboard interactions, allowing the system to efficiently monitor expiry dates and generate alerts without making the user interface responsible for the monitoring operations.



Components:

1. User Dashboard
2. Domain Monitoring Service
3. SSL Certificate Monitoring Service
4. Alert and Notification
5. Domain and Certificate Database

Interfaces:

1. Domain Monitoring API
2. SSL Monitoring API
3. Domain Expiry Data
4. Certificate Expiry Data
5. Alert Status Notification
6. Database Access